INCH-POUND

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01 September 2021
SUPERSEDING
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19 September 1986

DETAIL SPECIFICATION

CLOTH, KNITTED, RASCHEL, ARAMID

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers an aramid raschel knitted cloth.

….1.2 Classification. The cloth is available in one (1) type of aramid raschel knit. (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed: Attn: DLA Troop Support, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, you may want to verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil.

AMSC N/A FSC 8305

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-3064 - Evaluation of Quality of Textile Materials

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

FEDERAL TRADE COMMISSION

16 CFR Part 303 Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies of this document are available online at https://www.ftc.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP1 - Evaluation Procedure for Gray Scale for Color Change
AATCC EP2 - Evaluation Procedure for Gray Scale for Staining
AATCC EP9 - Evaluation Procedure for Visual Assessment of Color Difference of Textiles
AATCC TM16.3 - Test Method for Colorfastness to Light: Xenon Arc
AATCC TM20 - Test Method for Fiber Analysis: Qualitative
AATCC TM61 - Test Method for Colorfastness to Laundering: Accelerated
AATCC TM81 - Test Method for pH of the Water-Extract from Wet Processed Textiles

(Copies of these documents are available online at https://www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D3776/D3776M - Standard Test Methods for Mass per Unit Area (Weight) of Fabric
ASTM D5034 - Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)
ASTM D6413/D6413M - Standard Test Method for Flame Resistance of Textiles (Vertical Test)

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(Copies of these documents are available online at https://www.astm.org.)

SAE AMS-STD-595 – Colors Used in Government Procurement

(Copies of this document are available online at https://www.standards.sae.org/amsstd595/.

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations, unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Standard sample. Unless otherwise indicated, the finished cloth shall match the standard sample for shade and appearance, and shall, be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Material.

3.4.1 Yarn. The yarn shall be multifilament aramid (see 6.4). The yarn shall not char at a temperature less than 675ºF when tested as specified in 4.5.

3.4.2 Knit weave. The cloth shall be of the mesh type knitted on a single needle-bar raschel utilizing three (3) guide bars. Four ends per guide shall be used in knitting the cloth.

3.5 Color. The color of the finished cloth shall be Camouflage Green 483 matching SAE AMS-STD-595, color chip 34094 or as specified in the contract or purchase order. The color shall be obtained by the use of dry spun solution dyed fibers.

3.5.1 Visual shade matching. The color and appearance of the dyed cloth shall match the standard sample when tested as specified in 4.5.

3.5.2 Colorfastness. Unless otherwise specified in the contract or procurement documents, the finished dyed cloth shall conform to the colorfastness requirements listed below in Table I when tested as specified in 4.5.

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TABLE I. Colorfastness requirements.

	Laundering (Color Change and Staining) (3 cycles) (min.)	Light (after 6 AFU or 23 kJ/(m^2nm) @420 nm) (min.) 1/
Requirement	3-4	2-3

1/ AFU: AATCC Fading Units

3.6 Physical requirements. The finished cloth shall conform to the physical and flame resistance requirements listed below in Table II, when tested as specified in 4.5.

TABLE II. Physical requirements.

Characteristic	Requirement
Weight, oz./yd^2	
Minimum	7.0
Maximum	10
Meshes per square inch	
Minimum	50
Maximum	59
Breaking strength, pounds, (min.)	
Wales	200
Courses	115
Ultimate elongation, percent (max.)	
Wales	95
Courses	125
Flame Resistance:	
Char length, inches (max.)	
Wales	3.5
Courses	3.5
After flame, seconds, (max.)	
Wales	2.0
Courses	2.0
After glow, seconds, (max.)	
Wales	25.0
Courses	25.0
pH (range)	4.0-8.5

3.8. Finishing. The cloth shall be desized, scoured, and heat set. The cloth shall be heat set to meet the requirements of 3.8.1 (see 6.4).

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3.8.1 Curling. The finished cloth shall lie flat, without distortion and shall show no evidence of curling when tested as specified in 4.5.

3.9 Width. For Government procurements only, the width of the finished cloth shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of the selvage.

3.10 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put-up full width on a roll as specified in 5.1.

3.11 Fiber identification. Each roll of the finished cloth shall be labeled or ticketed for fiber content in accordance with the 16 CRF Part 303 Rules and Regulations Under the Textile Fiber Products Identification Act.

3.12 Workmanship. The finished cloth shall be uniform in quality and shall conform to the quality of product established on this specification. The occurrence of defects as specified in 4.3 and 4.4, shall not exceed the quality acceptance levels as specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance and finished defects listed in 4.4 and tested for the characteristics as specified in 4.5.

4.3 Conformance inspection. Conformance inspection shall include the visual examination of 4.4 and the tests of 4.5 through 4.7 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQLs) as specified in the contract or purchase order, except where otherwise indicated (see 6.2).

4.3.1 Inspection conditions. Unless otherwise specified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with this specification and all the requirements of referenced documents.

4.4 Visual examination. Each roll in the sample shall be examined yard by yard on the face side for defects in accordance with MIL-STD-3064, Type V.

4.4.1 Roll identification and marking examination. During the yard-by-yard examination each roll in the sample shall be examined for roll identification and marking defects as specified in MIL-STD-3064.

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4.4.2 Shade variation examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation as specified in MIL-STD-3064.

4.4.3 Length examination (individual and total yardage). During the yard-by-yard examination, each roll in the sample shall be examined for length as specified in MIL-STD-3064.

4.5 Material testing. The cloth shall be tested for the characteristics listed in Table III. The methods of testing as specified wherever applicable and as listed in Table III shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be two (2) continuous yards full width of the finished cloth for all physical and chemical tests. The lot shall be unacceptable if one (1) or more tests fail to meet the requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE III. Material tests.

Characteristic	Requirement Reference	Test Method
Fiber identification	See 3.4.1	AATCC TM20 (see 6.5)
Melting point	See 3.4.1	AATCC TM20 (see 6.5)
Weave	See 3.4.2	Visual 1/
Visual shade matching	See 3.5.1	See 4.6.1
Colorfastness: Laundering Light	Table I	 AATCC TM 61, Test 2A 2/ AATCC TM16.3, Opt. 3 3/
Weight	Table II	ASTM D3776/D3776M (Opt. C)
Meshes per square inch	Table II	Visual 1/
Breaking strength, lbs.	Table II	ASTM D5034
Ultimate elongation, percent	Table II	ASTM D5034
Flame resistance	Table II	ASTM D6413/D6413M
pH	Table II	AATCC TM81
Curling	See 3.8.1	See 4.6.2

1/ One (1) determination shall be made from each sample and results reported as “pass” or “fail”.
2/ Rated using the AATCC EP1, and AATCC EP2
3/ Rated using the AATCC EP1

4.6 Methods of testing and inspection.

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4.6.1 Visual shade matching. The color and appearance of the material shall match the standard sample when viewed using the AATCC EP9, Option C (see 6.6), with sources simulating artificial daylight D_{75} illuminant with a color temperature of 7500 (± 200) K illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2856 (± 200) K.

4.6.2 Curling. Two (2) specimens of cloth shall be cut 1-1/2-inches wide by 6-inches long, one (1) with the long dimension parallel to the wales direction and the other with the long dimension parallel to the courses direction. Both specimens shall be placed on a flat surface for at least five (5) minutes and then visually examined for evidence of curling.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD- ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cloth is intended for use in the manufacturing of survival vests.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. The specific issue of individual documents referenced (see 2.2)
c. Color required (see 3.5)
d. When first article is required (see 3.1, 4.2)
e. Width of cloth required (see 3.9)
f. Length required if other than specified (see 3.10)
g. Conformance inspection (see 4.3)
h. Inspection conditions (see 4.3.1)
i. Packaging (see 5.1)

6.3 Standard sample. For access to standard samples address the contracting activity issuing the invitation for bids or request for proposal.

6.4 Aramid. The requirements of the fabric can be met with Nomex® fiber, Type 432 or Type 433.

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6.5 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.6 Visual shade matching. In 2017, Option A of AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles was changed to Option C. NOTE: In case of confusion, the viewing geometry should be such that the specimen plane and illumination source are parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5°) angle relative to the horizontal. The observer will view the specimens at a 90° angle, relative to the plane of the specimens.

6.7 Changes from previous issue. Marginal notations are not used on this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.8 Subject term (key word) listing.

Curling
Mesh
Nomex

Custodians:
Army - GL
Navy - NU
Air Force - 11

Preparing Activity:
DLA-CT

(Project 8305-2021-019)

Review activities:
Army – MD

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at https://assist.dla.mil/.

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